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UPPER-BODY CONTEXT NODE — DERIVED FEATURES, QUALITY INDICES, ARTIFACT REJECTION, AND CONTEXT-GATED INTERPRETATION (NON-LIMITING)

A. Context Capture and Windowing (Non-Limiting)

In various embodiments, Node 5 continuously or periodically captures upper-body context signals and segments them into windows. A computing system selects windows for analysis and/or reporting based on validity gating. Non-limiting windows may be selected for rest, activity, sleep, recovery, or user-defined sessions.

B. Cardiac Features and HRV (Non-Limiting)

In various embodiments, Node 5 derives cardiac features from ECG and/or PPG, including: 1. heart rate (HR) and beat-to-beat variability; 2. time-domain HRV features (e.g., RMSSD, SDNN) and confidence indicators; 3. frequency-domain or spectral HRV features; and 4. arrhythmia plausibility checks and signal-quality gating. In certain embodiments, HRV features are computed only for windows meeting motion and contact integrity criteria.

C. Respiratory Dynamics (Non-Limiting)

In various embodiments, Node 5 derives respiratory dynamics using one or more modalities including chest motion patterns, inertial features, impedance-based respiration, or equivalent methods. Derived outputs may include: 1. respiration rate and variability; 2. respiration pattern stability indices; and 3. coupling confidence indicators. In certain embodiments, respiration outputs are suppressed or down-weighted during high-motion windows.

D. Autonomic Activation and EDA (Non-Limiting)

In various embodiments, Node 5 derives autonomic activation features including: 1. EDA/skin conductance tonic levels and phasic event rates; 2. activation indices computed from EDA and HRV combined features; and 3. stress/activation context tags with confidence indicators. In certain embodiments, the system computes a combined autonomic state score using confidence-weighted features.

E. Motion State and Artifact Rejection (Non-Limiting)

In various embodiments, the IMU supports activity classification and artifact rejection. Non-limiting derived features include: 1. motion magnitude and jerk; 2. posture/orientation stability; 3. step or movement detection; and 4. motion state tags (still, light movement, active, high-motion). In certain embodiments, high-motion windows are excluded from HRV and/or EDA reporting.

F. Thermal Context and Coupling Plausibility (Non-Limiting)

In various embodiments, temperature sensing is used to provide thermal context and coupling plausibility checks. Non-limiting uses include: 1. detecting plausible wear-state via temperature trends; 2. gating windows where thermal coupling indicates poor contact; and 3. correcting temperature-dependent drift for optical signals.

G. Contact Integrity and Wear Detection (Non-Limiting)

In various embodiments, contact integrity is verified using electrode impedance checks (for ECG), optical coupling checks (for PPG), capacitive proximity, or equivalent measures. The system may compute: 1. a wear integrity index; 2. a contact integrity index; and 3. a per-window confidence indicator.

H. Context-Gated Interpretation for Other Nodes (Non-Limiting)

In optional multi-node embodiments, Node 5 produces context tags that condition interpretation of other node signals. Non-limiting conditioning behaviors include: 1. excluding windows where Node 5 indicates high motion or poor wear integrity; 2. selecting matched windows (e.g., rest-state windows) for correlation across nodes; 3. applying context-specific baselines (e.g., rest vs. active normalization); and 4. confidence-weighting cross-device correlation scores using Node 5 quality indices.